ভারতীয় প্রযুক্তিবিদ্যা প্রতিষ্ঠান খড়্গপুর

भारतीय प्रौद्योगिकी संस्थान खड़गपुर

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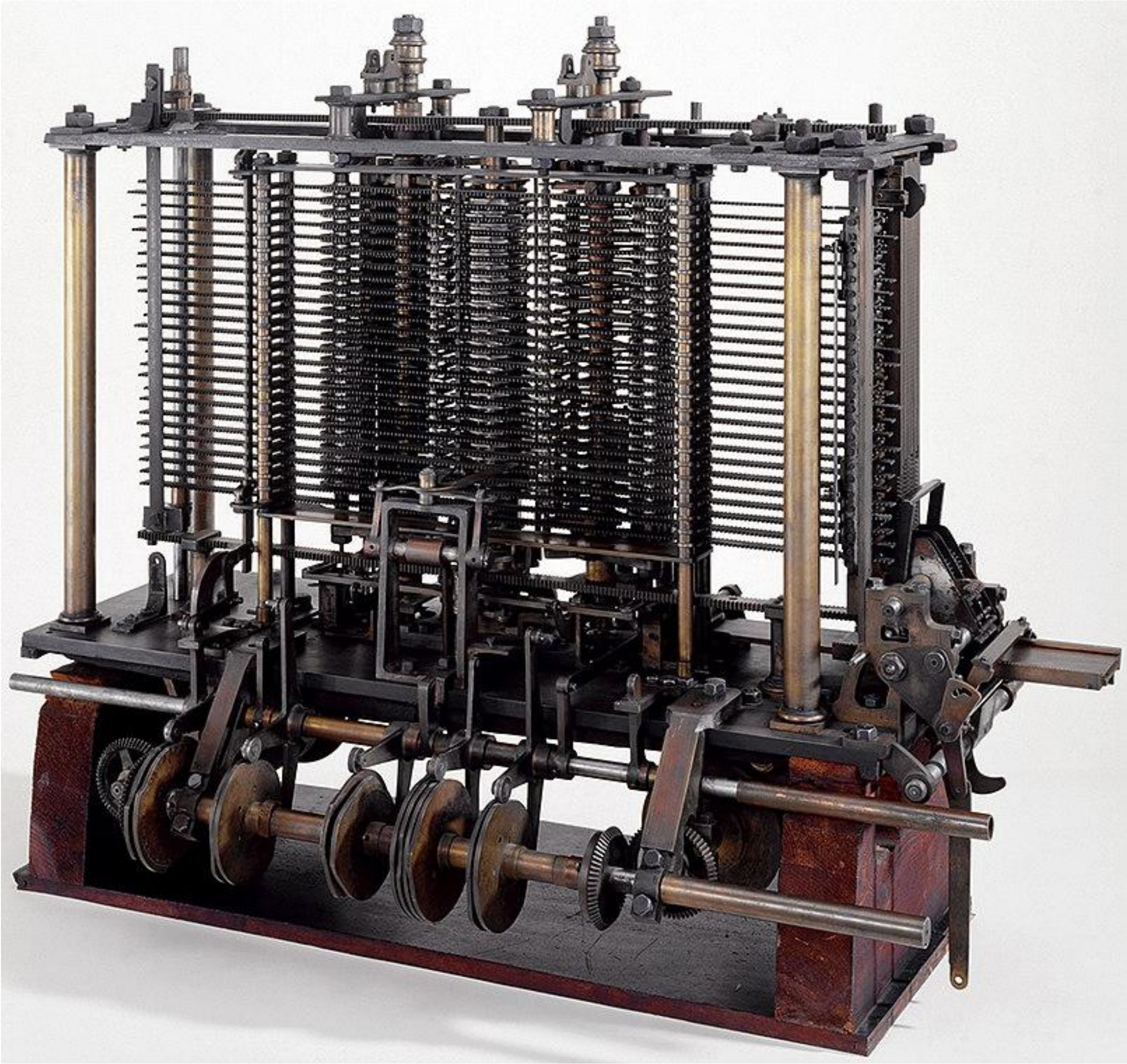
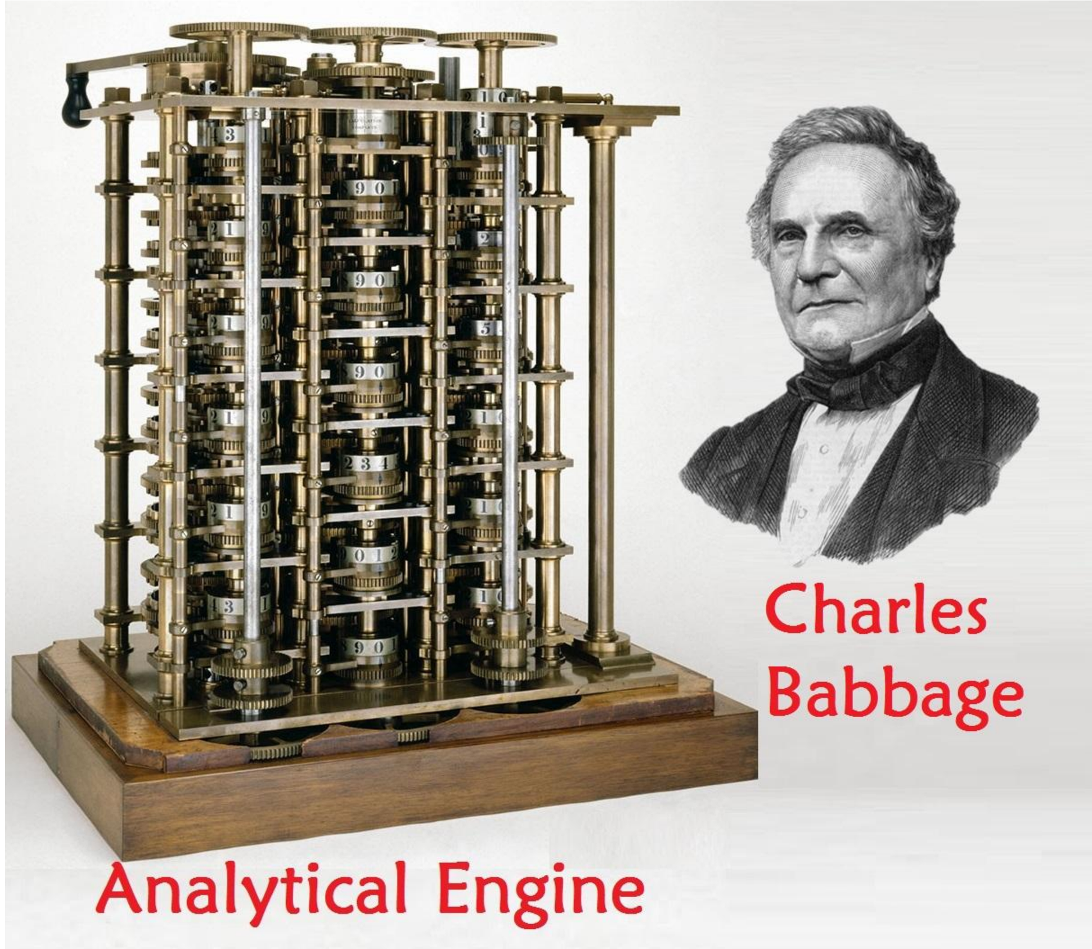
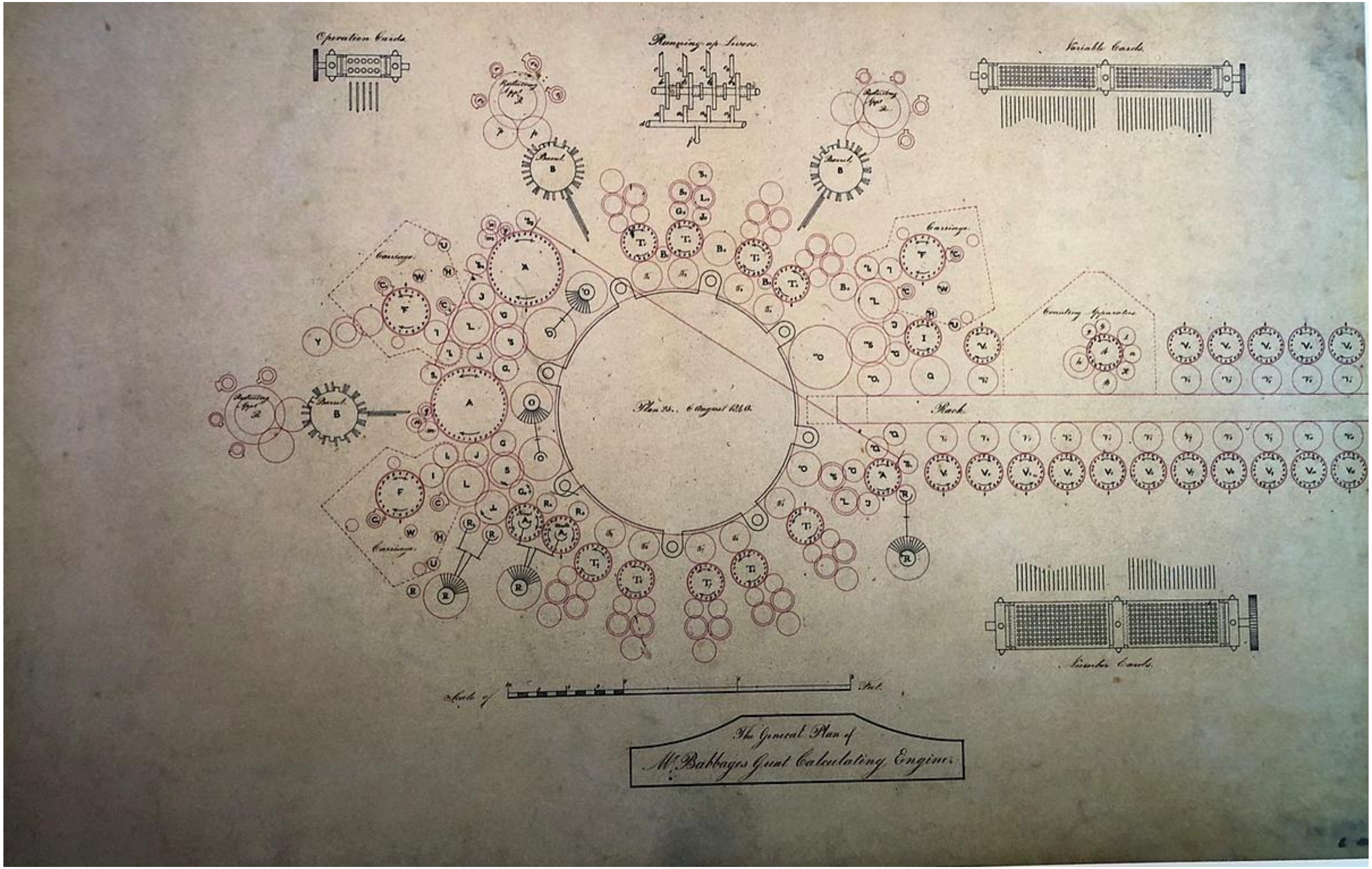
Indian Institute of Information Technology, Design and Manufacturing, Jabalpur

Analytical Engine 1837



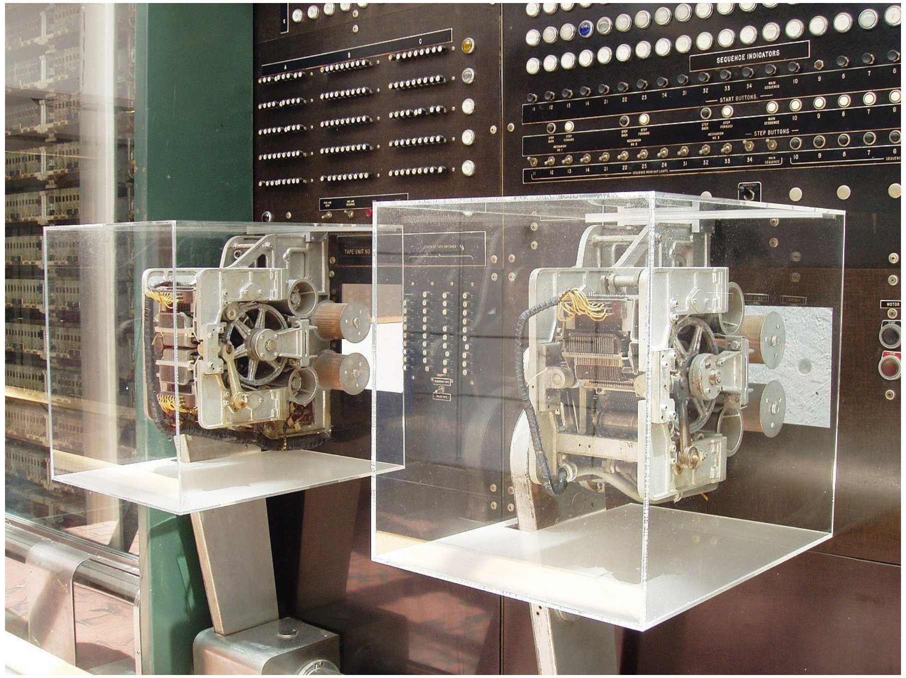




Plan diagram of the Analytical Engine from 1840

Harvard Mark-I, 1944

- IBM Automatic Sequence Controlled Calculator (ASCC)
- General-purpose electromechanical computer
- One of the first programs to run on the Mark I was initiated on 29 March 1944 [1] by John von Neumann.



Mainframe computers and Time Sharing

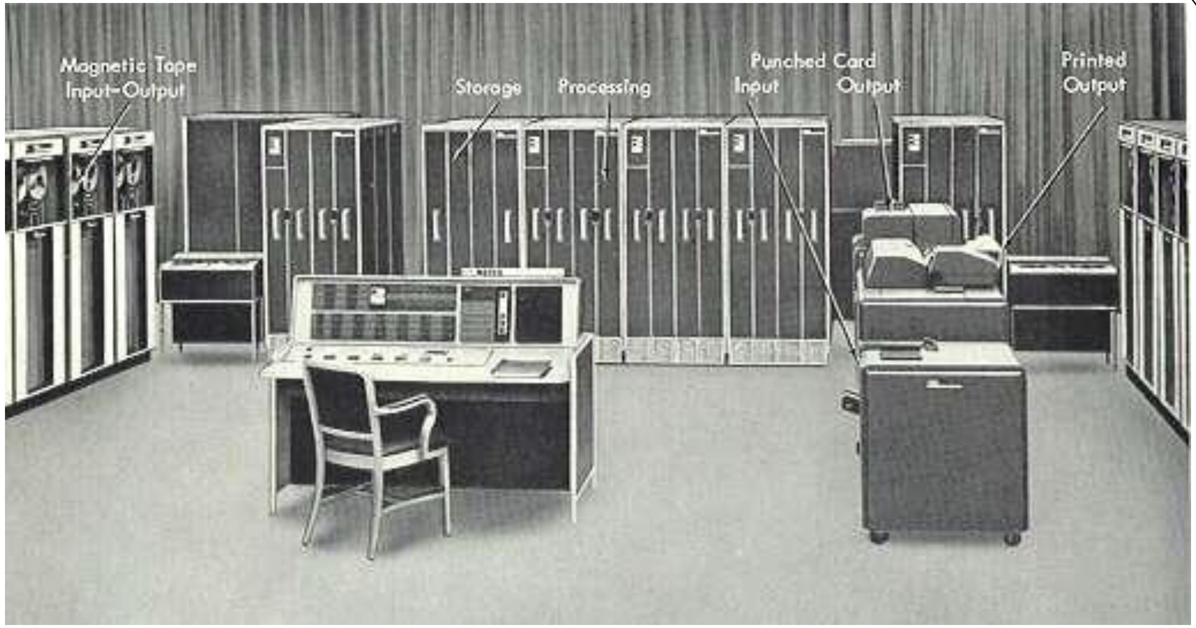
- **The 1950s**
  - large-scale mainframe computers
  - Practice of sharing CPU time on a mainframe became known in the industry as time-sharing.
- **Mid 70s**
  - time-sharing was popularly known as RJE (Remote Job Entry) by IBM and DEC.

http://en.wikipedia.org/wiki/Cloud_computing

Joe Thompson and Whirlwind, 1951 | MIT

- Among the first digital electronic computers that operated in real-time for output, and the first that was not simply an electronic replacement of older mechanical systems.
- One of the first computers to calculate in parallel (rather than serial), and was the first to use magnetic-core memory.

Timesharing (1960's)



- MIT IBM 7090 hardware
  - Transistorized version of IBM 709
  - 32K of 36-bit words of magnetic storage
  - 3 channels with 19 tape units
  - 60 cycle/sec accounting and interrupt clock
  - Special mode for memory protection, dynamic relocation and trapping of all user I/O
  - Time-sharing between 4 users (3 typewriters)
  - Single OS, a few simultaneous users
- Four significant design features of the foreground system for users:
  1. Enables development of programs in languages compatible with the background system
  2. Can develop a private file of programs
  3. Enables debugging sessions at the state of the previous session
  4. Setting their own pace without wasting computer time

IBM System/360 Model 67 (Aug'65)

- First IBM system with virtual memory capabilities
  - addition of the Dynamic Address Translation (DAT)
  - 16 MB (24-bit addresses)
  - Supported segmentation and paging
  - Memory virtualization enables multiple applications
  - Very high reliability/availability
  - Single OS, many simultaneous users

DEC PDP-11 (1970)

- 16-bit successor to the PDP-8
- UNIX provided virtualization (strong isolation) of memory, computation, storage
  - Single OS, many users
  - Low-cost alternative to mainframe-based computing, but less capable

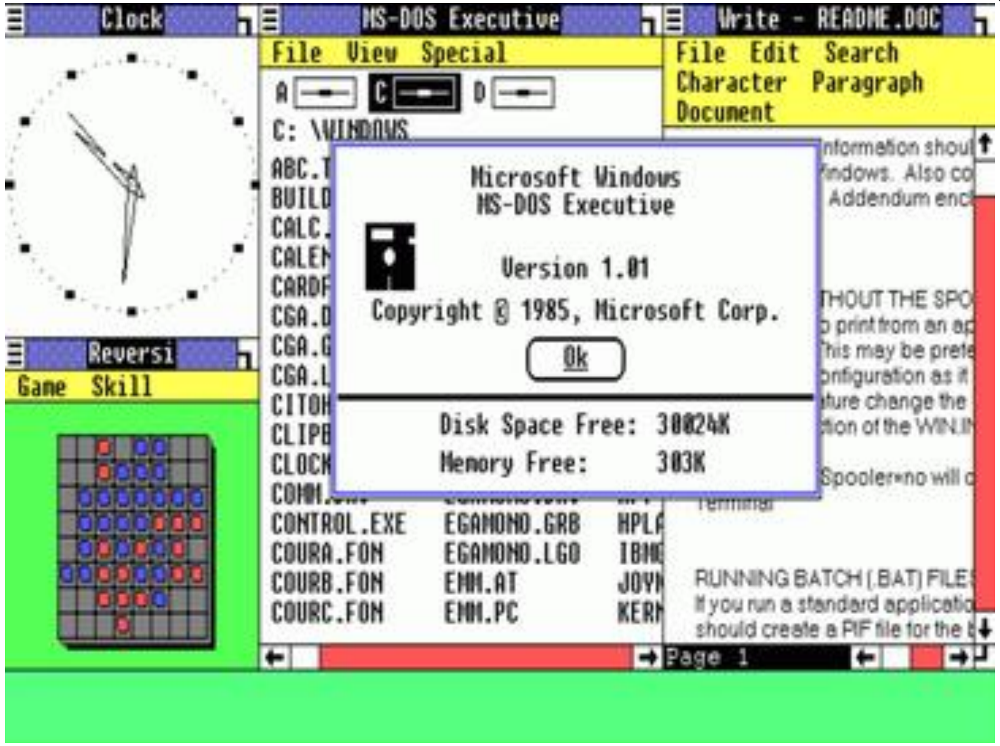
IBM System/370 (Aug'72)

- Virtual memory
  - 2KB or 4KB pages of memory,
  - 64KB or 1MB segment sizes
  - High performance Virtual Address translation
    - using "Translation-Lookaside Buffer" (TLB)
- VM/370 – Virtual Machine Manager (VMM)
  - First instance of hardware assisted virtualization
  - Privileged I/O: special-purpose control instructions, privileged I/O instructions
  - Privileged System Control: privileged system-control instructions
  - Complete virtualization (strong isolation) of memory, computation, storage
- *Multiple simultaneous OSes (VMs)*, 100's-1,000's simultaneous users
  - But very, very expensive platform

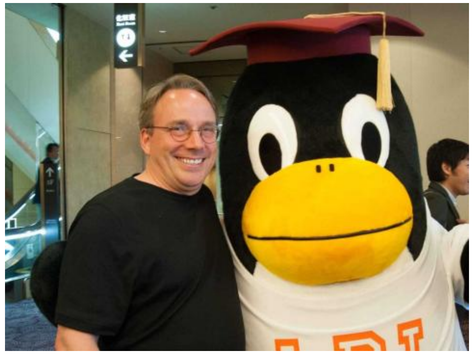
Personal computing era

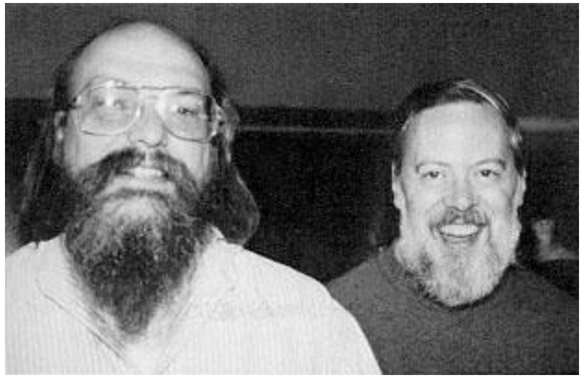
ATARI **micro**computers Gaming + home computing 1979.

Apple Macintosh, 1984



Microsoft Windows 1.0, 1985





Linux began in 1991, Linus Torvalds began a project that later became the Linux kernel. Based on Unix developed by Ken Thompson (left) and Dennis Ritchie (right), creators of the Unix operating system 1969, written in C and assembly language

Intel Pentium Processor, 1993

Internet search and services

Yahoo 1994

Google 1998

Operating System and Computer Network

➔ **Since 1990s**

- Telecommunications companies, provided point-to-point data circuits, Virtual Private Network (VPN).
- Efficient bandwidth.
- Big servers and network infrastructure.
- Scientists and technologists need for large-scale computing through time-sharing.
- Codes to optimize the infrastructure.
- Better OS platform, and applications to prioritize CPUs.
- **Origin of the term *Cloud*** is unclear.
  - ***Cloud*** was used as a metaphor for the ***Internet***.

http://en.wikipedia.org/wiki/Cloud_computing

Cloud Computing Definition

"A large-scale distributed computing paradigm that is driven by economies of scale, in which a pool of abstracted, virtualized, dynamically-scalable, managed computing power, storage, platforms, and services are delivered on demand to external customers over the Internet."

By Foster, Y. Zhau, R. Ioan, and S. Lu. "Cloud Computing and Grid Computing: 360-Degree Compared." *Grid Computing Environments Workshop*, 2008.

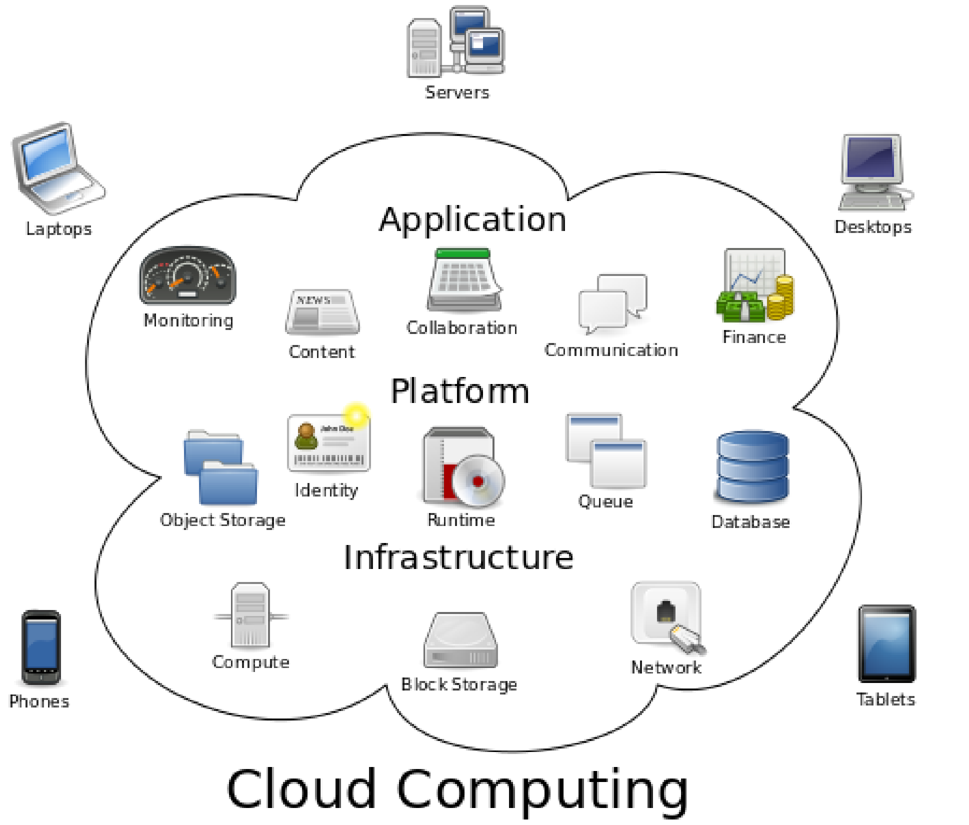
"A model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

By *National Institute of Standards and Technology (NIST)*, 2011.

Cloud Computing

- **Cloud computing** (Utility and Grid Computing).
- Clouds are of three types
  - public, private or hybrid.

Creating a groups of scalable hardware and software that allow remote computation, data storage and access using “Services”.



Whether cloud stands on existing technology?

Yes, Hypervisor, Virtualization, Load Balancing, VPN, SAN, SOA, Web Service and many more are the technologies behind Cloud.

Larry Ellison, CEO, Oracle (Wall Street Journal, Sept. 26, 2008) "we have redefined Cloud Computing to include everything that we already do.... change the wording of some of our ads."

Richard Stallman, Founder, Free Software Foundation (The Guardian, Sept. 29, 2008). *said* "it's marketing hype. Somebody is saying this is inevitable ..... it's very likely to be a set of businesses campaigning to make it true."

Mell, Peter, and Tim Grance. "The NIST definition of cloud computing. "*National Institute of Standards and Technology* 53.6 (2009): 50.

Historical Evolution of Terminologies

Supercomputers

Grid Computing

Utility Computing

- pay-as-you-go computing
- Illusion of infinite resources
- Demanded billing like hourly, daily, number of users

Cloud Computing

(Infrastructure, Hardware, Platform, Software) as a service → "X as a service"

Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS)

Requirements of CAPEX and OPEX

Cloud computing is required to meet the requirement of scalability, cost efficiency, legal agreement and business.

CAPEX and OPEX

- CAPEX model (Capital expenditure)
  - buy the dedicated hardware over a period of time.
- OPEX model (Operational expenditure)
  - use a shared cloud infrastructure and pay as one uses it.
- **Moving to cloud:** CAPEX ➔➔➔ OPEX model
- **Results:** Cloud vendors are experiencing high growth rates annually
- Business in profit.

CAPEX to OPEX

- **Benefit of OPEX and why cloud?**
  1. Sharing of resources to achieve coherence and economies of scale.
  2. Avoid upfront infrastructure costs
  3. Focus on code development instead of infrastructure.
  4. Applications can run faster, with improved manageability and less maintenance, and enables IT.
  5. Adjustment with fluctuating and unpredictable business demand.
  6. Cloud providers use "pay as you go" model. Charges may be high if administrators do not used cloud.
  7. Reason for the growth of cloud computing: availability of high-capacity networks, low-cost computers and storage devices, hardware virtualization, service-oriented architecture, utility computing.

Requirements of Cloud Computing

- Cloud computing in computer science is analogous to electricity grid over an electric network.
- Concept of converged infrastructure and shared services.
- Maximizing the effectiveness of the shared resources.
- Resources are shared by multiple users.
- Resources are dynamically reallocated as per demand.
- Efficient use of computing power thus being eco-friendly (less power, air conditioning, rackspace, etc.).
- Multiple users can work on a machine.
- Users can retrieve and update the data with flexible licenses for different applications.

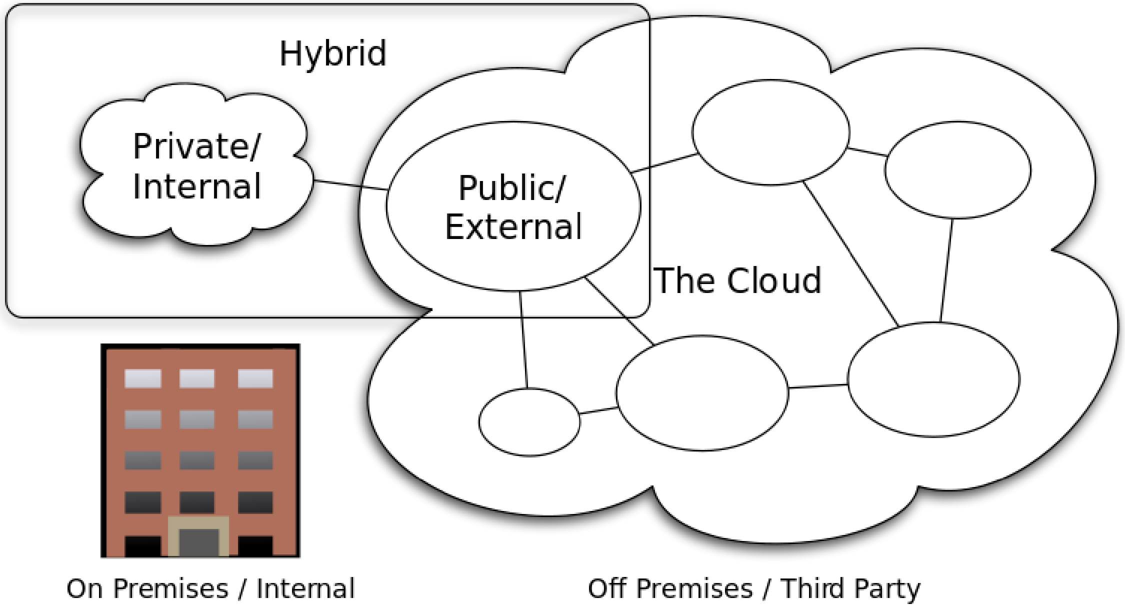
Technologies involved in Cloud

- SaaS
- Inexpensive storage
- Inexpensive and plentiful client CPU bandwidth to support significant client computation
- Sophisticated client algorithms, including HTML, CSS, AJAX, REST
- Client broadband
- SOA (service-oriented architectures)
- Large infrastructure implementations from Google, Yahoo, Amazon, and others that provided real-world, massively scalable, distributed computing
- Commercial virtualization
- Cloud computing is upgraded version kind of grid computing
- Cloud has evolved by addressing the QoS (quality of service) and reliability problems.
- Cloud computing provides the tools and technologies to build data/compute intensive parallel applications.

Creeger, Mache. "Cloud Computing: An Overview." *ACM Queue* 7.5 (2009): 2.

https://sites.google.com/site/animeshchaturvedi07/academic-teaching/cloudcomputing

Private, Public and Hybrid Cloud



Cloud Computing Types

Private, Public and Hybrid Cloud

- **Private cloud:** "The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units). It may be owned, managed, and operated by the organization, a third party, or some combination of them, and it may exist on or off premises".
- **Public cloud:** "The cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider".
- **Hybrid cloud:** "The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities, but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds)".

The NIST Definition of Cloud Computing

Public cloud

- Public cloud services may be free or offered on a **pay-per-usage** model.
- Mostly similar public and private cloud architecture. But difference is for security, storage, and other resources that are usually better in public cloud.
- Amazon AWS, Microsoft and Google operate the infrastructure at their data center.
- **Pay-as-you-go** model for compute and "Intelligence-as-a-Service."
- Default for training massive AI models (LLMs) due to unrivaled GPU access.
- **Global Hyperscale:** Amazon (AWS), Microsoft (Azure), and Google (GCP) dominate, followed by emerging "NeoClouds".
- **Neoclouds** are specialized AI cloud infrastructure providers of high-performance GPU-as-a-Service (GPUaaS) to meet the massive demand for AI training and inference.

Private cloud

- Private cloud establishment requires effort by an organization to and also intelligence to take decisions regarding the resources.
- It improves business and also raises security issues that must be addressed.
- Private data centers are generally capital intensive.
- Requires space, hardware, environmental controls and electricity.
- Operational expenses as well as **Capital expenditures** for renewable.
- Issues "buy, build, manage and expertise”.

Hybrid cloud

- Hybrid cloud will help in extension of the capacity or the capability of a cloud service, by aggregation, integration or customization with another cloud service. Ability to connect, managed or dedicated services with both private and public cloud resources.
  1. If an organization have some client data on a private cloud application, but want to interconnect that application to a business intelligence application provided on a public cloud as a software service.
  2. If an organizations use public cloud computing resources to meet temporary capacity needs that can not be met by the private cloud.
- **Cloud bursting:** when the demand for computing capacity increases then an application deployment model in a private cloud or data center will "bursts" to a public cloud. During spike in processing demand requirement private cloud infrastructure that supports average workloads will start using cloud resources from public or private clouds together.
- **Advantage:** payment only for the extra compute resources when they are needed.

Hybrid Cloud

- AWS and Microsoft offers direct connect services.
- **Dedicated Interconnects:** High-speed and dedicated private links like
  - **AWS Direct Connect:** connections between on-premises data center to AWS
  - **Microsoft Azure ExpressRoute:** on-premises infrastructure and Azure datacenters
  - are now essential for low-latency Hybrid-AI workflows, often bypassing the public internet entirely for security.
  - High-bandwidth connectivity essential for large data transfers, real-time analytics, and hybrid cloud environments.
- **Architectural:** Public and Private clouds share "Cloud-Native" architectures (Kubernetes/Containers).
- Difference is **Data Sovereignty** and **Zero-Trust control** in Private clouds versus the infinite **GPU-elasticity** of Public clouds.

Elasticity and Provisioning

Architecture for Elasticity

- **Vertical Scale-Up**
  - Keep on adding resources to a unit to increase computation power.
  - Process the job to single computation unit with high resources.
- **Horizontal Scale-Out**
  - Keep on adding discrete resources for computation and make them behave as in converged unit.
  - Splitting job on multiple discrete machines, combine the output.
  - Distribute database.
- For HPC second option is better than first. Because Complexity and cost of first option is very high.

Cloud Computing Infrastructure

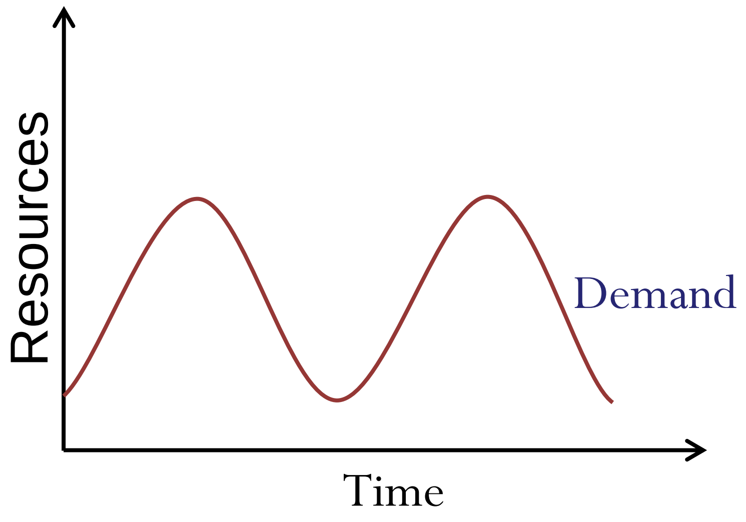
- Scale "out", not scale "up". This is due to limitations of
  - Serial Management Protocol (SMP) and
  - Large shared-memory machines
- Move processing to the data
  - Cluster has limited bandwidth
- Process data sequentially, avoid random access
  - Seeks are expensive, disk throughput is reasonable
- Seamless scalability for ordinary programmers
  - From the mythical man-month to the tradable machine-hour

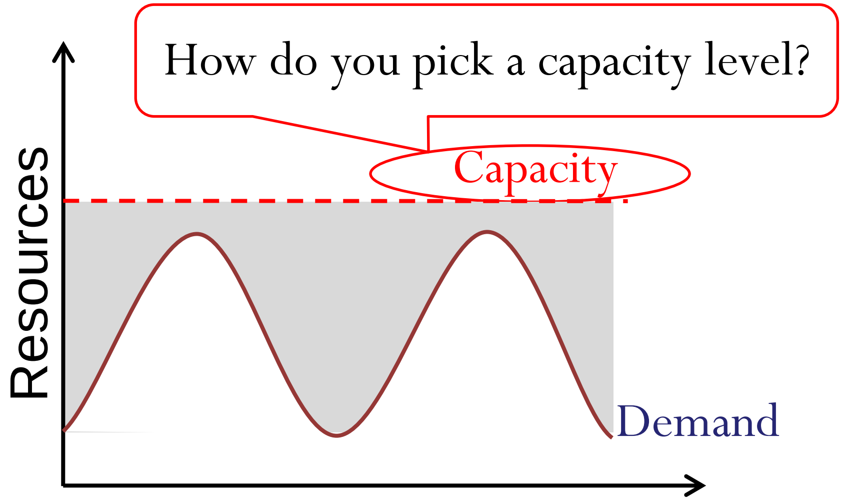
Cloud Computing Computation Models

- Finding the right level of abstraction
  - von Neumann architecture vs cloud environment
- Hide system-level details from the developers
  - No more race conditions, lock contention, etc.
- Separating the *what* from *how*
  - Developer specifies the computation that needs to be performed
  - Execution framework (“runtime”) handles actual execution
- Idempotent operations of API (if we make multiple identical requests with the same input parameters and receive the same response every time)
  - Simplifies redo in the presence of failures

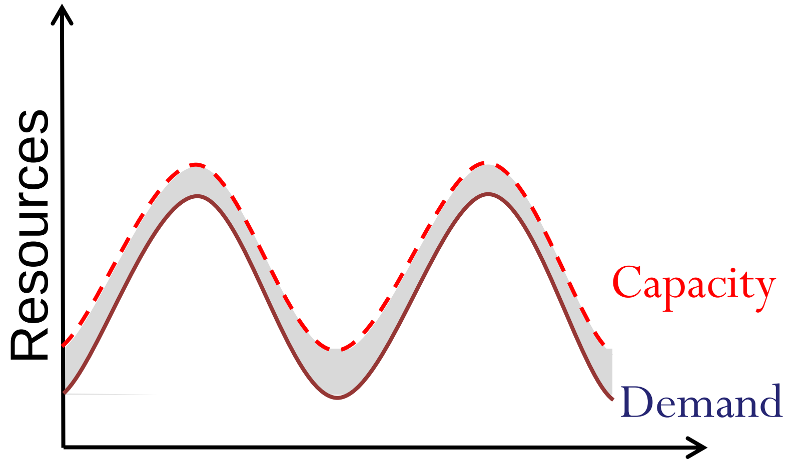
Cloud Application Demand

- Many cloud applications have cyclical demand curves
  - Daily, weekly, monthly, …
- Workload spikes more frequent and significant
- Economics of Cloud Users
  - Pay by use instead of provisioning for peak





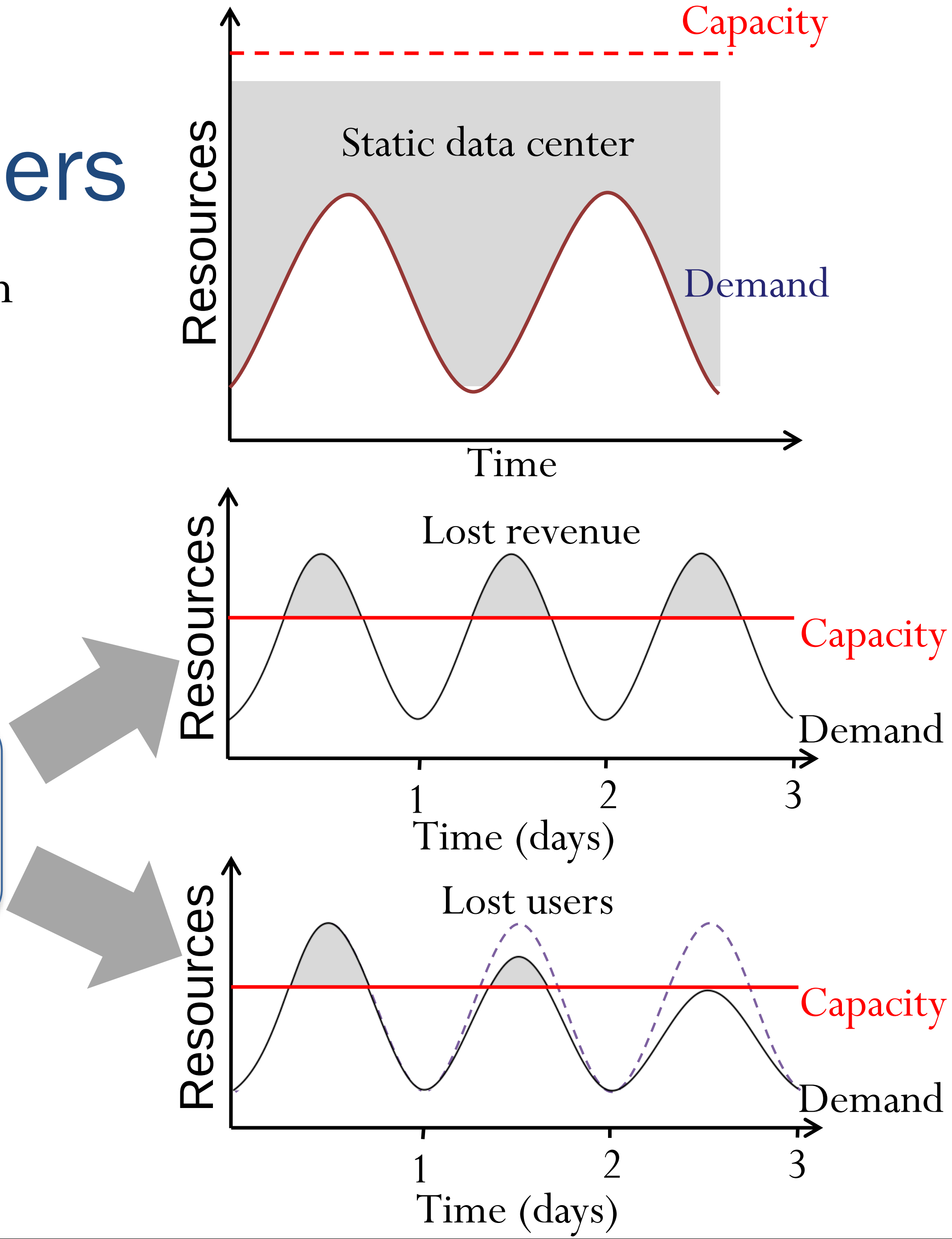
Static data center



Data center in the cloud

Economics of Cloud Users

- Risk of over-provisioning: underutilization
- Huge sunk cost in infrastructure
- Heavy penalty for under-provisioning



Risk of underutilization if peak predictions are too optimistic
– Wasted CapEx
Hard to provision for spiky workloads

Unused resources

Resource Management & Provisioning

Public cloud services can be used with three Cloud provisioning:

- **Consumer self-provisioning:** Consumer contract and pay as per usage for cloud services directly to provider,
  - e.g. Institute Google or Microsoft domain (email, form, docx, excel etc.).
- **Advanced provisioning:** Consumer contract and pay in advance for resources and services
  - e.g. online event management system
- **Dynamic provisioning:** Provider allocates resources as per consumer usage, then de-provisioning when resources are not in use.
  - Consumer pays as per usage
- **Provisioning and orchestration:**
  - create, modify, and delete resources
  - orchestrate workflows and management of workloads

https://en.wikipedia.org/wiki/Cloud_management

Elasticity and Resource Provisioning

Elasticity: provisioning and de-provisioning resources in an autonomic manner,

For Elasticity **Avoid**

- Over-provisioning: allocating more resources than required,
  - Issue: pay for the useless resources
- Under-provisioning: allocating fewer resources than required
  - Issue: poor service performance, e.g. slow or unreachable
  - Issue: loses customers

https://en.wikipedia.org/wiki/Elasticity_(cloud_computing)

Data Centers
(Hardware & software Infrastructure)

Data Centers

- A **data center** is a facility used to house computer systems and associated components, such as telecommunications and storage systems.
- Includes backup power supplies, redundant data communications connections, environmental controls (e.g., air conditioning, fire suppression) and various security devices.
- Large data centers are industrial scale operations using as much electricity as a small town.

Scale of Cloud in industries?

10s of CPUs?

100s of CPUs?

Scale of Cloud in industries?

- Example: Large scale log processing
  - Clusters vary from 1 to thousands of nodes.
  - Thousands of CPUs and computers
- Cluster has hundreds to thousands of nodes
  - Each node
    - A×B CPU boxes
    - A×B TB disk
    - X GB RAM

"Cabinet Asile" by Robert.Harker - Own work. Licensed under CC BY-SA 3.0 via Wikimedia Commons

Hyperscale Industry Metrics

- **Massive Cluster Growth:** Modern AI workloads now run on clusters exceeding **65,000 to 130,000 nodes** in leading GCP/Azure environments.

- **Compute Density:** Individual nodes have evolved significantly:
  - **CPUs:** 64 -128+ cores per socket.
  - **RAM:** 1TB - 2TB+ per node is common for high-memory AI inference.
  - **Accelerators:** 8x NVIDIA H100/H200 or Microsoft Maia GPUs per rack.

- **Storage:** Shift from 1TB disks to high-speed **NVMe SSDs (Non-volatile Memory Express Solid State Drives)** (4TB–15TB per drive) to feed data to GPUs at wire speed.

Data Centers Energy

- **The Power Shift:** Modern data centers are **Industrial-Scale Gigawatt (GW) Campuses**. By 2026, global consumption is expected to exceed **1,000 TWh**, equivalent to the annual energy use of Japan.
- **Sustainable Infrastructure:** Focus on **Liquid Cooling** (direct-to-chip or immersion) to handle high-density AI racks (60kW–100kW+ per rack).
- **Energy Mix:** Leading providers now use **Small Modular Reactors (SMRs)** or hydrogen fuel cells to ensure 24/7 carbon-free power.
- **Advanced Controls:** AI-driven "Digital Twins" manage heat flow and workload distribution in real-time.

Virtualization & Services

- Cloud computing is a combination of existing technologies. User can use this combination without the technical knowledge or expertise.
- **Virtualization:** It is a process of creating a illusion of a physical computing device, into one or more low-configuration virtual device, such that they can easily be used and managed.
- **Service-oriented Architecture (SOA):** Cloud computing adopts concepts of SOA that to create services for sub-problems and these services can be integrated to solution the problem. Moreover, these services are further divided into individual operations (procedures).
- Cloud computing provides its resources in the form of services. Use the well-established technology domains like SOA and Virtualization. This results in globalized and user-friendly technology.

Other relevant technologies

- Mainframe computer — Powerful computers used for critical applications for bulk data processing.
- Peer-to-peer — Distributed Architecture without central coordination. Participants are both suppliers and consumers of resources (in contrast to the traditional client–server model).
- Grid computing — computer are composed into a cluster using networked, cluster is acting as converged resource.
- Utility computing —The package of computing resources (CPU and storage) are charged according to the pay-as-you-go.
- Service oriented Computing: This organize and utilize the distributed services offered by different owners. It gives a formal way of offering, discover, interaction with the flexibility to orchestrate according to requirement.

References

1. Fox Armando, Rean Griffith, A. Joseph, R. Katz, A. Konwinski, G. Lee, D. Patterson, A. Rabkin, and I. Stoica. "Above the clouds: A Berkeley view of cloud computing." Dept. Electrical Eng. and Comput. Sciences, University of California, Berkeley, Rep. UCB/EECS 28 (2009): 13.
2. Creeger, Mache. "Cloud Computing: An Overview." ACM Queue 7.5 (2009): 2.
3. http://www.w3.org/TR/2004/NOTE-ws-gloss-20040211
4. http://en.wikipedia.org/wiki/Web_Services_Description_Language,
5. http://en.wikipedia.org/wiki/Cloud_computing
6. http://aws.amazon.com/what-is-cloud-computing/
7. http://en.wikipedia.org/wiki/Data_center
8. https://www.microsoft.com/en-us/research/blog/leslie-lamport-receives-turing-award/
9. https://en.wikipedia.org/wiki/Hypervisor

תודה רבה
Hebrew

Ευχαριστώ
Greek

Спасибо
Russian

Danke
German

धन्यवादः
Sanskrit

Merci
French

ধন্যবাদ
Bangla

நன்றி
Tamil

شكراً
Arabic

ಧನ್ಯವಾದಗಳು
Kannada

Thank You
English

നന്ദി
Malayalam

Grazie
Italian

多謝
Traditional Chinese

ధన్యవాదాలు
Telugu

આભાર
Gujarati

ਧੰਨਵਾਦ
Punjabi

धन्यवाद
Hindi & Marathi

Gracias
Spanish

多谢
Simplified Chinese

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Obrigado
Portuguese

ありがとうございました
Japanese

ขอบคุณ
Thai

감사합니다
Korean